

AN EXPLICIT DEGREE- m MINIMAL POLYNOMIAL: THE FACTORISATION HALF OF CONJECTURE 11 OF LASJAUNIAS

ABSTRACT. Let p be an odd prime, $k \geq 1$ and $m \geq 2$ integers with $p = km + 1$, and let $V = [p, k, k, \varepsilon_A]$ and $H(V) \in \mathbb{F}_p[T][X]$ be as in Lasjaunias, *J. Integer Seq.* **21** (2018), Article 18.8.3. Conjecture 11 of that paper asserts that $H(V)$ has an irreducible factor P of degree m in X , vanishing at the hyperquadratic root α of its Proposition 9, with irrationality measure $\nu(\alpha) = m$, and given by the explicit binomial $P(X) = \sum_{0 \leq i \leq m} (-1)^{ki} \binom{m}{i} T^{(1-(-1)^i)/2} X^{m-i}$. We prove, for every admissible triple (p, k, m) , that this P is irreducible of degree m over $\mathbb{F}_p(T)$, that it is the minimal polynomial of α , and that it divides $H(V)$ in $\mathbb{F}_p[T][X]$; along the way, $\varepsilon_A = (-1)^k$. The layer $m = 2$ is the last clause of that paper's Proposition 9 and the case $k = 1$ is its printed example at $p = 3$. The remaining clause $\nu(\alpha) = m$ of the conjecture is the source's own Corollary 5 and is not our result. Of Proposition 9, which we do not reprove, the proof uses only that some irrational $\alpha \in \mathbb{F}_p((1/T))$ with a pole at $T = \infty$ satisfies $H(V)(\alpha) = 0$; the accompanying program constructs such an α outright, with initial partial quotient $w_{k+1}T$, for every admissible triple with $p \leq 60$.

1. THE CONJECTURE

Throughout, p is an odd prime, $\mathbb{F}_p$ the field with p elements, and T an indeterminate. We work in $\mathbb{F}_p(T)$ and in its completion $\mathbb{F}_p((1/T))$ at the place $T = \infty$.

Lasjaunias [1] attaches to a quadruple $V = [p, k, j, \varepsilon]$ with $1 \leq k < p/2$, $1 \leq j \leq 2k - 1$ and $\varepsilon \in \mathbb{F}_p^*$ a polynomial $H(V)$ of degree $p + 1$ in X , and studies its factorisation. The following is Conjecture 11 of [1], quoted verbatim from the e-print [2], where it is numbered Conjecture 5.3.

Let $k \geq 1$ and $m \geq 2$ be integers. Let p be a prime number such that $p = km + 1$. Let $V = [p, k, k, \varepsilon_A]$ and $H(V)$ be defined as above. Then the polynomial $H(V)$ has a factor P in $\mathbb{F}_p[T][X]$ such that $P(\alpha) = 0$, $\deg_X(P) = m$ and $\nu(\alpha) = m$. More precisely, we have

$$(1) \quad P(X) = \sum_{0 \leq i \leq m} (-1)^{ki} \binom{m}{i} T^{(1-(-1)^i)/2} X^{m-i}.$$

2020 *Mathematics Subject Classification.* 11J70, 11T55, 12E05.

Key words and phrases. hyperquadratic continued fraction, power series field, odd characteristic, minimal polynomial, continuant, irreducibility.

The journal wording is strictly stronger — *an irreducible factor P* — with the closed form (1) character-for-character identical. Theorem 1 below covers that stronger form.

Theorem 1. *Let $k \geq 1$ and $m \geq 2$ be integers with $p = km + 1$ prime, let $V = [p, k, k, \varepsilon_A]$, and let $\alpha \in \mathbb{F}_p((1/T))$ be any element irrational over $\mathbb{F}_p(T)$ with $H(V)(\alpha) = 0$ and with a pole at $T = \infty$; Proposition 9 of [1] supplies one, whose initial partial quotient is $w_{k+1}T$, and for $p \leq 60$ `verify.py` constructs such an α — one with that same initial partial quotient — outright. Put $c = (-1)^k T$. Then*

- (i) $\varepsilon_A = (-1)^k$;
- (ii) *the polynomial P of (1) satisfies $2P(X) = (1+c)(X+1)^m + (1-c)(X-1)^m$, is monic of degree m in X , and is irreducible over $\mathbb{F}_p(T)$;*
- (iii) $P(\alpha) = 0$; *consequently α has degree exactly m over $\mathbb{F}_p(T)$, and P is its minimal polynomial;*
- (iv) P divides $H(V)$ in $\mathbb{F}_p[T][X]$.

Parts (ii)–(iv) are the factorisation half of Conjecture 11, in the stronger journal form. The third clause $\nu(\alpha) = m$ of the conjecture, where ν is the irrationality *measure* and not the degree, is *not* proved here and is not ours: it is Corollary 5 of [1], which gives $\nu(\alpha) = 2 + (p - 2k - 1)/\ell$ with $\ell = 2k - j$, and at $j = k$, $p = km + 1$ this is $2 + m - 2 = m$ identically. Combining Theorem 1 with that corollary closes Conjecture 11 as printed.

Two cases of the factorisation half were already in print. The layer $m = 2$, i.e. $k = (p - 1)/2$, is the last clause of Proposition 9 of [1], and the case $k = 1$ is the printed example at $p = 3$.

Remark. Clause (iii) is about a root α of $H(V)$ in $\mathbb{F}_p((1/T))$ with a pole at $T = \infty$, and that pole is exactly what selects $\zeta = 1$ in its proof. Only the *existence* of such an α is quoted, from Proposition 9 of [1] as paraphrased in (F4) of Section 2; no other property of it is used, so the reading of (F4) cannot influence what is proved. The pole is not a further reading either: the first partial quotient of α is by definition its polynomial part $[\alpha]$, and $[\alpha] = w_{k+1}T$ with $w_{k+1} \neq 0$ says exactly that $v(\alpha) = -1 < 0$. For every admissible triple with $p \leq 60$ the program of Section 4 supplies the existence itself rather than quoting it: it constructs α as a Laurent series, with $[\alpha] = w_{k+1}T$, certifies $H(V)(\alpha) = 0$ to order T^{-20} , and exhibits the one consequence of irrationality the proof uses, namely the relation $\alpha^p = (A\alpha + B)/(B\alpha + A)$ obtained by solving $H(V)(\alpha) = 0$. It also certifies the Hensel inequality $v(H(V)(\alpha)) > 2v(H(V)'(\alpha))$, which upgrades the truncated identity to an exact root of $H(V)$ in the complete field $\mathbb{F}_p((1/T))$.

2. NOTATION

We transcribe the definitions of [1]. Set

$$\omega_k = (-1)^{k-1} \prod_{1 \leq i \leq k} \left(1 - \frac{1}{2i}\right) \in \mathbb{F}_p^*, \quad w_1 = (2k - 1)(2k \omega_k)^{-1},$$

and define $w_2, \dots, w_{2k} \in \mathbb{F}_p^*$ by $w_{i+1}w_i = (2k-2i-1)(2k-2i+1)(i(2k-i))^{-1}$ for $1 \leq i \leq 2k-1$. For $1 \leq n \leq r \leq 2k$ let $K_{n,r} = \langle w_n T, \dots, w_r T \rangle \in \mathbb{F}_p[T]$ be the continuant of the displayed word, with the conventions $K_{n,n-1} = 1$ and $K_{n,n-2} = 0$; thus $K_{n,r} = w_r T K_{n,r-1} + K_{n,r-2}$, $\deg K_{n,r} = r - n + 1$ and $\text{lc } K_{n,r} = \prod_{n \leq i \leq r} w_i$. We write $\langle w_n, \dots, w_r \rangle = K_{n,r}(1)$ for the continuant of the bare word, and $[a_1, \dots, a_n] = a_1 + 1/(a_2 + 1/(\dots))$ for a continued-fraction value in $\mathbb{F}_p$. Finally

$$H([p, k, j, \varepsilon]) = K_{j+2,2k} X^{p+1} - K_{j+1,2k} X^p + \varepsilon(K_{1,j} X + K_{1,j-1}),$$

$$\varepsilon_A = (-1)^{k+j+1}[w_{j+1}, \dots, w_{2k}, \omega_k], \quad \varepsilon_B = (-1)^{k+j+1}[w_{j+1}, \dots, w_{2k}].$$

The bracket in ε_A is a continued-fraction *value*, not a continuant.

The proof in Section 3 uses the following four facts from [1], as paraphrased here, and nothing else from it.

- (F1) (eq. (2)) $w_{2k+1-i} = -w_i$ for $1 \leq i \leq 2k$.
- (F2) (eq. (4)) $K_{1,2k} = (-1)^k(T^2 - 1)^k$ and $K_{2,2k} = (-1)^k R_k$, where R_k is the remainder of T^p on division by $(T^2 - 1)^k$.
- (F3) (eq. (5)) $K_{n,r} K_{t,r-1} - K_{n,r-1} K_{t,r} = (-1)^{r-t} K_{n,t-2}$ for $t \leq r$.
- (F4) (Proposition 9) For $V = [p, k, j, \varepsilon_A]$ there is $\alpha \in \mathbb{F}_p((1/T))$ with $H(V)(\alpha) = 0$, irrational over $\mathbb{F}_p(T)$, whose continued-fraction expansion in $\mathbb{F}_p((1/T))$ begins with the partial quotient $w_{j+1}T$; in particular α has a pole at $T = \infty$. This is our wording, not the source's sentence; the proof below invokes it only for the two properties just displayed, namely $H(V)(\alpha) = 0$ and $v(\alpha) < 0$, and `verify.py` exhibits an α with both (and with $[\alpha] = w_{j+1}T$, $j = k$) for every admissible triple with $p \leq 60$.

Equation (6) of [1], $K_{k+1,2k} = (-1)^k K_{1,k}$ and $K_{k+2,2k} = (-1)^{k-1} K_{1,k-1}$, is the special case of Lemma 2 below that we use most, and we reprove it there rather than quote it.

For the rest of the paper we fix $k \geq 1$, $m \geq 2$ with $p = km + 1$ prime, and take $j = k$; this is admissible because $k = (p-1)/m \leq (p-1)/2 < p/2$. We abbreviate

$$c = (-1)^k T, \quad a = \frac{c-1}{c+1} \in \mathbb{F}_p(T), \quad P_k = (T^2 - 1)^k,$$

$$g_r = K_{1,r}(1), \quad h_r = K_{2,r}(1) \quad (r \geq 0).$$

3. PROOF

Lemma 2 (reversal). *For $1 \leq n \leq r+1 \leq 2k+1$ we have $K_{n,r} = (-1)^{r-n+1} K_{2k+1-r, 2k+1-n}$, and the same identity holds for the bare continuants $\langle w_n, \dots, w_r \rangle$.*

Proof. A continuant is unchanged by reversing its word. Reversing $(w_n T, \dots, w_r T)$ gives $(w_r T, \dots, w_n T)$, which by (F1) equals $(-w_{2k+1-r} T, \dots, -w_{2k+1-n} T)$. Negating all s entries of a word multiplies its continuant by $(-1)^s$: if $L_i = (-1)^i K_i$ then $L_i = (-a_i) L_{i-1} + L_{i-2}$ with $L_{-1} = 0$, $L_0 = 1$. Here

$s = r - n + 1$. Nothing in the argument uses the factor T , so it applies verbatim to the bare word. $\square$

The four instances we use are

$$(2) \quad \begin{aligned} K_{k+1,2k} &= (-1)^k K_{1,k}, & K_{k+2,2k} &= (-1)^{k-1} K_{1,k-1}, \\ K_{k+1,2k-1} &= (-1)^{k-1} K_{2,k}, & K_{k+2,2k-1} &= (-1)^{k-2} K_{2,k-1}; \end{aligned}$$

the first two are eq. (6) of [1]. We also use the concatenation rule $K_{n,r} = K_{n,i}K_{i+1,r} + K_{n,i-1}K_{i+2,r}$ for $n \leq i \leq r$, which is the associativity of the product of the matrices $\begin{pmatrix} w_i T & 1 \\ 1 & 0 \end{pmatrix}$.

Lemma 3. (a) $K_{1,k}^2 - K_{1,k-1}^2 = P_k$. (b) $K_{1,k}K_{2,k} - K_{1,k-1}K_{2,k-1} = R_k$. (c) $P_k(K_{2,k}^2 - K_{2,k-1}^2) = R_k^2 - 1$.

Proof. Concatenate at $i = k$ and substitute (2). With $(n, r) = (1, 2k)$, $K_{1,2k} = K_{1,k}K_{k+1,2k} + K_{1,k-1}K_{k+2,2k} = (-1)^k(K_{1,k}^2 - K_{1,k-1}^2)$, and (F2) gives (a). With $(n, r) = (2, 2k)$, $K_{2,2k} = (-1)^k(K_{2,k}K_{1,k} - K_{2,k-1}K_{1,k-1})$, and (F2) gives (b). With $(n, r) = (2, 2k-1)$, $K_{2,2k-1} = (-1)^{k-1}(K_{2,k}^2 - K_{2,k-1}^2)$. On the other hand Lemma 2 at $(n, r) = (1, 2k-1)$ gives $K_{1,2k-1} = -K_{2,2k}$, and (F3) at $(n, t, r) = (1, 2, 2k)$ gives $K_{1,2k}K_{2,2k-1} - K_{1,2k-1}K_{2,2k} = (-1)^{2k-2}K_{1,0} = 1$; hence $(-1)^k P_k K_{2,2k-1} + R_k^2 = 1$, i.e. $K_{2,2k-1} = (-1)^k(1 - R_k^2)/P_k$. Comparing the two expressions for $K_{2,2k-1}$ gives (c). $\square$

Lemma 4. $g_k^2 = g_{k-1}^2$, $g_k h_k - g_{k-1} h_{k-1} = 1$, and $h_k^2 - h_{k-1}^2 = -2 \binom{(p-1)/2}{k} = 2\omega_k$ in $\mathbb{F}_p$.

Proof. Put $T = 1$ in Lemma 3(a): $P_k(1) = 0$. Since $T^p - R_k$ is divisible by P_k and $P_k(1) = 0$, we get $R_k(1) = 1$; putting $T = 1$ in Lemma 3(b) gives the second relation.

For the third, write $z = T^2 - 1$ and $u = T - 1$, and let $Q_k = (T^p - R_k)/P_k \in \mathbb{F}_p[T]$. Since p is odd,

$$T^p = T(T^2)^{(p-1)/2} = T(1+z)^{(p-1)/2} = T \sum_{0 \leq i \leq (p-1)/2} \binom{(p-1)/2}{i} z^i,$$

and as $\deg z^i = 2i$ the polynomial part after division by z^k is $Q_k = T \sum_{i \geq k} \binom{(p-1)/2}{i} z^{i-k}$, so

$$(2) \quad Q_k(1) = \binom{(p-1)/2}{k}.$$

Next, $T^p = (1+u)^p = 1+u^p$ in characteristic p , so $1 - R_k = 1 - T^p + P_k Q_k = -u^p + u^k(T+1)^k Q_k = u^k[(T+1)^k Q_k - u^{p-k}]$, and $p - k = k(m-1) \geq 1$; hence $(1 - R_k)/u^k$ takes the value $2^k Q_k(1)$ at $T = 1$. Since $(1 + R_k)(1) = 2$ and $(T+1)^k|_{T=1} = 2^k$,

$$\left. \frac{1 - R_k^2}{P_k} \right|_{T=1} = \frac{1 - R_k}{(T-1)^k} \cdot \frac{1 + R_k}{(T+1)^k} \Big|_{T=1} = 2^k Q_k(1) \cdot \frac{2}{2^k} = 2Q_k(1).$$

By Lemma 3(c), $h_k^2 - h_{k-1}^2 = (R_k^2 - 1)/P_k|_{T=1} = -2Q_k(1)$, which with (3) is the first stated value. Finally, since $k < p$ the factorial $k!$ is invertible in $\mathbb{F}_p$ and $\binom{x}{k}$ is a polynomial in x over $\mathbb{F}_p$; at $x = (p-1)/2 = -1/2$,

$$\binom{-1/2}{k} = \frac{1}{k!} \prod_{0 \leq i \leq k-1} \left(-\frac{1}{2} - i\right) = \frac{(-1)^k}{k!} \prod_{0 \leq i \leq k-1} \frac{2i+1}{2} = (-1)^k \frac{(2k)!}{4^k (k!)^2},$$

i.e. $\binom{-1/2}{k} = (-1)^k \binom{2k}{k} 4^{-k}$, while $\omega_k = (-1)^{k-1} \prod_{1 \leq i \leq k} \frac{2i-1}{2i} = (-1)^{k-1} \binom{2k}{k} 4^{-k}$. Therefore $-2\binom{(p-1)/2}{k} = 2(-1)^{k-1} \binom{2k}{k} 4^{-k} = 2\omega_k$. $\square$

Lemma 5. $g_{k-1} \neq 0$, $g_k = (-1)^{k+1} g_{k-1}$, and $\varepsilon_A = (-1)^k$.

Proof. By Lemma 4 there is $s \in \{1, -1\}$ with $g_k = s g_{k-1}$; substituting into $g_k h_k - g_{k-1} h_{k-1} = 1$ gives

$$(4) \quad g_{k-1} (s h_k - h_{k-1}) = 1,$$

so $g_{k-1} \neq 0$ and $s h_k \neq h_{k-1}$.

Suppose $k \geq 2$. Then (F3) at $(n, t, r) = (1, 2, k)$ reads $K_{1,k} K_{2,k-1} - K_{1,k-1} K_{2,k} = (-1)^{k-2} K_{1,0} = (-1)^k$, so at $T = 1$, $g_{k-1} (s h_{k-1} - h_k) = (-1)^k$. Dividing by (4), $(s h_{k-1} - h_k) = (-1)^k (s h_k - h_{k-1})$. For $s = 1$ the left side is the negative of the right factor, forcing $(-1)^k = -1$; for $s = -1$ it equals it, forcing $(-1)^k = 1$. Hence $s = (-1)^{k+1}$. For $k = 1$ one has $\omega_1 = 1/2$, $w_1 = 1$, so $g_1 = w_1 = 1 = g_0$ and again $s = 1 = (-1)^{k+1}$.

Now $\varepsilon_A = (-1)^{2k+1} [w_{k+1}, \dots, w_{2k}, \omega_k] = -[w_{k+1}, \dots, w_{2k}, \omega_k]$. Expressing the continued-fraction value as a ratio of bare continuants, $[a_1, \dots, a_n] = \langle a_1, \dots, a_n \rangle / \langle a_2, \dots, a_n \rangle$, and applying Lemma 2 to the four bare continuants involved,

$$\begin{aligned} \langle w_{k+1}, \dots, w_{2k}, \omega_k \rangle &= \omega_k (-1)^k g_k + (-1)^{k-1} h_k = (-1)^k (\omega_k g_k - h_k), \\ \langle w_{k+2}, \dots, w_{2k}, \omega_k \rangle &= \omega_k (-1)^{k-1} g_{k-1} + (-1)^{k-2} h_{k-1} = (-1)^{k-1} (\omega_k g_{k-1} - h_{k-1}), \end{aligned}$$

whence

$$(5) \quad \varepsilon_A = \frac{\omega_k g_k - h_k}{\omega_k g_{k-1} - h_{k-1}}.$$

Suppose first that k is even, so $s = -1$ and (4) reads $g_{k-1} (h_k + h_{k-1}) = -1$; in particular $h_k + h_{k-1} \neq 0$. Then, using Lemma 4,

$$2\omega_k g_{k-1} = (h_k^2 - h_{k-1}^2) g_{k-1} = (h_k - h_{k-1}) \cdot g_{k-1} (h_k + h_{k-1}) = h_{k-1} - h_k,$$

so the numerator of (5) is $-\omega_k g_{k-1} - h_k = (\omega_k g_{k-1} - h_{k-1}) - (2\omega_k g_{k-1} + h_k - h_{k-1}) = \omega_k g_{k-1} - h_{k-1}$, i.e. numerator and denominator agree. They are nonzero: if both vanished then $-\omega_k g_{k-1} = h_k$ and $\omega_k g_{k-1} = h_{k-1}$, so $h_k + h_{k-1} = 0$, excluded. Hence $\varepsilon_A = 1 = (-1)^k$. If instead k is odd then $s = 1$, (4) reads $g_{k-1} (h_k - h_{k-1}) = 1$, so $h_k \neq h_{k-1}$ and $2\omega_k g_{k-1} = (h_k + h_{k-1}) g_{k-1} (h_k - h_{k-1}) = h_k + h_{k-1}$; the numerator of (5) is then $\omega_k g_{k-1} - h_k = -(\omega_k g_{k-1} - h_{k-1}) + (2\omega_k g_{k-1} - h_k - h_{k-1}) = -(\omega_k g_{k-1} - h_{k-1})$, and again the denominator is nonzero, since otherwise $h_k = h_{k-1}$. Hence $\varepsilon_A = -1 = (-1)^k$. $\square$

Lemma 6. Set $U = K_{1,k} + K_{1,k-1}$ and $W = K_{1,k} - K_{1,k-1}$, and let $L = \prod_{1 \leq i \leq k} w_i$. Then $L^2 = 1$, and $\{U, W\} = \{L(T-1)^k, L(T+1)^k\}$ with $U = L(T-1)^k$ for k even and $W = L(T-1)^k$ for k odd. In either case

$$\lambda := \frac{K_{1,k} + K_{1,k-1}}{K_{1,k} - K_{1,k-1}} = a^k, \quad a = \frac{c-1}{c+1}, \quad c = (-1)^k T.$$

Proof. $\deg K_{1,k} = k$ and $\deg K_{1,k-1} = k-1$, so U and W both have degree k and leading coefficient L . By Lemma 3(a), $UW = (T-1)^k(T+1)^k$; comparing leading coefficients, $L^2 = 1$. For $k \geq 2$, (F3) at $(n, t, r) = (1, 2, k)$ exhibits $(-1)^k$ as an $\mathbb{F}_p[T]$ -combination of $K_{1,k}$ and $K_{1,k-1}$, so these are coprime, hence $\gcd(U, W)$ divides $\gcd(2K_{1,k}, 2K_{1,k-1}) = 2$, a unit; for $k = 1$ we have $U = T+1$ and $W = T-1$ outright. So U and W are coprime with product $(T-1)^k(T+1)^k$, and each is L times one of the two prime powers.

Which is which is decided at $T = 1$: by Lemma 5, $U(1) = ((-1)^{k+1} + 1)g_{k-1}$ and $W(1) = ((-1)^{k+1} - 1)g_{k-1}$ with $g_{k-1} \neq 0$, so for k even $U(1) = 0 \neq W(1)$ and for k odd $W(1) = 0 \neq U(1)$. If k is even then $c = T$, $U = L(T-1)^k$, $W = L(T+1)^k$ and $\lambda = ((T-1)/(T+1))^k = a^k$. If k is odd then $c = -T$, so $a = (-T-1)/(-T+1) = (T+1)/(T-1)$, while $U = L(T+1)^k$ and $W = L(T-1)^k$; again $\lambda = ((T+1)/(T-1))^k = a^k$. $\square$

Lemma 7. $2P(X) = (1+c)(X+1)^m + (1-c)(X-1)^m$, with P as in (1) and $c = (-1)^k T$. In particular P is monic of degree m in X .

Proof. $(1+c)(X+1)^m + (1-c)(X-1)^m = \sum_i \binom{m}{i} X^{m-i} [(1+c) + (1-c)(-1)^i]$, and the bracket is 2 for i even and $2c$ for i odd, i.e. $2c^{(1-(-1)^i)/2}$ in both cases. Since $c^{(1-(-1)^i)/2} = (-1)^{ki} T^{(1-(-1)^i)/2}$ — both sides being 1 for i even and $(-1)^k T$ for i odd — the sum equals $2P(X)$. The coefficient of X^m is $\binom{m}{0} = 1$. $\square$

Lemma 8. Let F be a field carrying a discrete valuation $v: F^* \rightarrow \mathbb{Z}$, let $b \in F^*$ and $m \geq 1$ with $\gcd(v(b), m) = 1$. Then $Y^m - b$ is irreducible over F .

Proof. Let β_0 be a root in an algebraic closure, $E = F(\beta_0)$ and $d = [E : F] \leq m$. Extend v to a valuation $\tilde{v}$ of E ; its value group is $\frac{1}{e}\mathbb{Z}$ where $e \leq d$ is the ramification index. From $m\tilde{v}(\beta_0) = v(b)$ we get $\tilde{v}(\beta_0) = v(b)/m$, so $ev(b)/m \in \mathbb{Z}$; as $\gcd(v(b), m) = 1$ this forces $m \mid e$. Then $m \leq e \leq d \leq m$, so $d = m$ and $Y^m - b$, being of degree m with a root of degree m , is irreducible. $\square$

Proof of Theorem 1. (i) is Lemma 5.

The relation, diagonalised. Write $\varepsilon = \varepsilon_A = (-1)^k$. By (F4), $H(V)(\alpha) = 0$, i.e. $K_{k+2,2k}\alpha^{p+1} - K_{k+1,2k}\alpha^p + \varepsilon(K_{1,k}\alpha + K_{1,k-1}) = 0$. Since α is irrational and $K_{1,k} \neq 0$, we may solve:

$$\alpha^p = \frac{A\alpha + B}{C\alpha + D}, \quad A = -\varepsilon K_{1,k}, \quad B = -\varepsilon K_{1,k-1}, \quad C = K_{k+2,2k}, \quad D = -K_{k+1,2k}.$$

By the reversal instances (2) and $\varepsilon = (-1)^k$ we get $C = (-1)^{k-1}K_{1,k-1} = -\varepsilon K_{1,k-1} = B$ and $D = -(-1)^k K_{1,k} = -\varepsilon K_{1,k} = A$. This is the point of the choice $j = k$: at that index the antisymmetry (F1) makes the Möbius matrix symmetric under $\begin{pmatrix} A & B \\ C & D \end{pmatrix} \mapsto \begin{pmatrix} D & C \\ B & A \end{pmatrix}$.

Put $\beta = (\alpha + 1)/(\alpha - 1)$, legitimate and nonzero because $\alpha \notin \mathbb{F}_p(T)$. Raising to the p -th power is additive, so $\beta^p = (\alpha^p + 1)/(\alpha^p - 1)$, and substituting $\alpha^p = (A\alpha + B)/(B\alpha + A)$,

$$\alpha^p + 1 = \frac{(A + B)(\alpha + 1)}{B\alpha + A}, \quad \alpha^p - 1 = \frac{(A - B)(\alpha - 1)}{B\alpha + A},$$

so

$$\beta^p = \lambda\beta, \quad \lambda = \frac{A + B}{A - B},$$

where $A - B = -\varepsilon(K_{1,k} - K_{1,k-1}) \neq 0$ by Lemma 6. Cancelling ε , $\lambda = (K_{1,k} + K_{1,k-1})/(K_{1,k} - K_{1,k-1}) = a^k$ by Lemma 6.

From $\beta^{p-1} = a^k$ to $\beta^m = a$. Since $\beta \neq 0$ we may divide: $\beta^{p-1} = a^k$, and $p - 1 = km$, so $(\beta^m)^k = a^k$, i.e. $(\beta^m/a)^k = 1$. As $k \mid p - 1$, the polynomial $Y^k - 1$ splits in $\mathbb{F}_p$, so $\beta^m = \zeta a$ for some $\zeta \in \mathbb{F}_p^*$ with $\zeta^k = 1$.

To identify ζ , work in $\mathbb{F}_p((1/T))$, whose valuation ring is $\mathbb{F}_p[[1/T]]$ with maximal ideal $\mathfrak{m} = (1/T)$. By (F4), α has a pole at $T = \infty$, hence $2/(\alpha - 1) \in \mathfrak{m}$ and $\beta = 1 + 2/(\alpha - 1) \equiv 1 \pmod{\mathfrak{m}}$; likewise $a = 1 - 2/(c + 1) \equiv 1$. Reducing $\beta^m = \zeta a$ modulo $\mathfrak{m}$ gives $1 = \zeta$. Therefore

$$(6) \quad \beta^m = a, \quad \text{i.e.} \quad (c + 1)(\alpha + 1)^m = (c - 1)(\alpha - 1)^m,$$

that is, $(1 + c)(\alpha + 1)^m + (1 - c)(\alpha - 1)^m = 0$, which by Lemma 7 is $2P(\alpha) = 0$. As p is odd, $P(\alpha) = 0$: clause (iii) up to the degree statement.

Irreducibility. Let $v = v_{T=1}$ be the valuation of $\mathbb{F}_p(T)$ at $T = 1$, normalised onto $\mathbb{Z}$. Then $v(c - 1)$ and $v(c + 1)$ are 1 and 0 for k even and 0 and 1 for k odd, so $v(a) = \pm 1$, coprime to m . By Lemma 8, $Y^m - a$ is irreducible over $\mathbb{F}_p(T)$, so each of its roots β_0 satisfies $[\mathbb{F}_p(T)(\beta_0) : \mathbb{F}_p(T)] = m$. Now $a \neq 1$, so $\beta_0 \neq 1$, and $\alpha_0 := (\beta_0 + 1)/(\beta_0 - 1)$ is defined and satisfies $\mathbb{F}_p(T)(\alpha_0) = \mathbb{F}_p(T)(\beta_0)$, because $X \mapsto (X + 1)/(X - 1)$ is an involution of $\mathbb{P}^1$ over $\mathbb{F}_p(T)$ and $\beta_0 = (\alpha_0 + 1)/(\alpha_0 - 1)$. Reversing the computation (6), $\beta_0^m = a$ is equivalent to $P(\alpha_0) = 0$, and this correspondence is a bijection between the roots of $Y^m - a$ and those of P ; it is defined in both directions because $a \neq 1$, so $\beta_0 \neq 1$, and $2P(1) = (1 + c)2^m \neq 0$ by Lemma 7, so no root of P equals 1. Hence every root of P has degree m over $\mathbb{F}_p(T)$; since $\deg_X P = m$ by Lemma 7, P is irreducible. This is (ii).

Conclusion. P is monic, irreducible of degree m , and $P(\alpha) = 0$, so P is the minimal polynomial of α and $\deg \alpha = m$; this completes (iii). Finally $P \mid H(V)$ in $\mathbb{F}_p(T)[X]$, and since P is monic in X the Euclidean division of $H(V) \in \mathbb{F}_p[T][X]$ by P takes place in $\mathbb{F}_p[T][X]$ and has the same quotient, so $P \mid H(V)$ in $\mathbb{F}_p[T][X]$: this is (iv). $\square$

4. VERIFICATION

No step of Theorem 1 rests on a machine computation. The program `verify.py`, whose recorded run is `verify.output.txt`, re-derives the objects displayed below from the definitions of Section 2 and re-runs the identities of Lemmas 2–7, together with the divisibility $P \mid H(V)$, on all 453 admissible triples $(p, (p-1)/m, m)$ with p an odd prime at most 300; it reports all 81 of its checks passing.

On the 71 of those triples with $p \leq 60$ it also *reconstructs* α . Newton's iteration for $H(V)$ in $\mathbb{F}_p((1/T))$, started at $\alpha_0 = w_{k+1}T$, converges because $H(V)(x) = e H(V)'(\alpha) + e^p(K_{k+2,2k}\alpha - K_{k+1,2k}) + K_{k+2,2k}e^{p+1}$ identically for $e = x - \alpha$, where $H(V)' = \partial_X H(V) = K_{k+2,2k}X^p + \varepsilon K_{1,k}$, so that v of the error is multiplied by p and shifted by $p+k-1-v(K_{k+2,2k}\alpha - K_{k+1,2k}) > 0$ at each step; every coefficient is exact in $\mathbb{F}_p$ and each series carries the exponent below which it is known, so no identity is asserted beyond the precision that was actually computed. The program then certifies, to order T^{-20} : $H(V)(\alpha) = 0$, with the Hensel inequality $v(H(V)(\alpha)) > 2v(H(V)'(\alpha))$ that makes the constructed series an exact root; $\lfloor \alpha \rfloor = w_{k+1}T$, hence the pole at $T = \infty$; the diagonalised relation $\alpha^p = (A\alpha + B)/(B\alpha + A)$; $\beta \equiv a \equiv 1 \pmod{1/T}$ and $\beta^p = a^k\beta$; the selection $\beta^m = a$, i.e. $\zeta = 1$; $P(\alpha) = 0$, both for the recomputed P and for the P printed below; and that *none* of the $k-1$ sibling solutions $\beta^m = \zeta a$, $\zeta^k = 1$, $\zeta \neq 1$, vanishes at α — which is what exact division alone cannot see. Three controls fail as they must: α moved by T^{-3} , a wrong initial partial quotient, and the ε_B form of H at α .

One consequence of the proof is worth recording, because it pins the initial partial quotient of α to the data of the conjecture: $\beta = 1 + 2/(\alpha - 1) = 1 + 2/(w_{k+1}T) + O(T^{-2})$ while $a = 1 - 2(-1)^k/T + O(T^{-2})$, so $\beta^m = a$ forces $2m/w_{k+1} = -2(-1)^k$, that is

$$w_{k+1} = (-1)^{k+1}m \quad \text{in } \mathbb{F}_p,$$

and the expansion of α begins with the partial quotient $(-1)^{k+1}mT$. The word recurrence of Section 2 reproduces this on all 453 triples of the census.

What remains cited rather than recomputed: Capelli's criterion (only its hypothesis $v_{T-1}(a) = \pm 1$ is re-derived), Hensel's lemma, the continued-fraction construction of Proposition 9 itself, the clause $\nu(\alpha) = m$, and every p beyond the census.

At the cell $(p, k, m) = (17, 4, 4)$ all objects are over $\mathbb{F}_p = \mathbb{F}_{17}$, with representatives in $\{0, \dots, 16\}$, and

$$\omega_4 = 15, \quad (w_1, \dots, w_8) = (7, 8, 6, 4, 13, 11, 9, 10), \quad \varepsilon_A = 1, \quad \varepsilon_B = 16,$$

$$K_{1,3} = 13T^3 + 13T, \quad K_{1,4} = T^4 + 6T^2 + 1, \quad K_{5,8} = T^4 + 6T^2 + 1, \quad K_{6,8} = 4T^3 + 4T.$$

The two polynomials below are printed in a plain normal form; whitespace and line breaks are not significant.

$$\begin{aligned} H = & (4T^3+4T)X^{18} + (16T^4+11T^2+16)X^{17} \\ & + (T^4+6T^2+1)X + (13T^3+13T) \end{aligned}$$

$$P = X^4 + 4TX^3 + 6X^2 + 4TX + 1$$

Here $H = H([17, 4, 4, 1])$ and P is (1) at $(k, m) = (4, 4)$. The quotient H/P has degree $14 = p+1-m$ in X ; it is recorded in `verify.py`, which multiplies it by P to recover H .

REFERENCES

- [1] A. Lasjaunias, *On particular families of hyperquadratic continued fractions in power series fields of odd characteristic*, J. Integer Seq. **21** (2018), Article 18.8.3. cs.uwaterloo.ca/journals/JIS/VOL21/Lasjaunias/lasj5.html. Statement labels of the form “Conjecture 11”, “Proposition 9”, “Corollary 5” are those of this version of record.
- [2] A. Lasjaunias, same title, e-print, arXiv:1803.01739. The e-print renumbers the statements, so “Conjecture 5.3”, “Proposition 5.1” and “Corollary 4.2” are its labels for Conjecture 11, Proposition 9 and Corollary 5 of [1], and the equation numbers (2), (4), (5), (6) quoted above are its numbers.